App Dev Project Report

1. Student Details

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About Me: I am interested in full-stack web development and database-backed applications.

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2. Project Details

Project Title: *Placement Portal Application*

Problem Statement:

Institutes need an efficient system to manage campus recruitment involving Admin (Placement Cell), Companies, Students, placement drives, and applications. Manual methods make approvals, tracking, and reporting difficult.

Approach:

The project is implemented using Flask + Jinja2 + Bootstrap + SQLite.
The application uses role-based flows for Admin, Company, and Student.
The database is created programmatically using SQLAlchemy models and initialization code (no manual DB creation).

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3. AI/LLM Declaration

I used AI assistance (ChatGPT/Codex) for guidance in code structuring, validation logic refinement, and documentation drafting.
Final integration, debugging, milestone-wise implementation, and project decisions were completed and verified locally.

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4. Technologies and Frameworks Used

Technolog / Library	Purpose

Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data
WERKZEUG SECURITY	Password hashing and verification

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5. Database Schema / ER Diagram

Tables:

admins : pre-seeded admin account (username/email/password_hash, active status)
companies: company profile and approval state (company_name, industry, hr_contact, email, approval flags)
students: student profile (roll_number, email, phone, degree, branch, cgpa, skills, resume)
placement_drives: drives posted by companies (title, eligibility, skills, deadline, status)
applications: student applications per drive (status, date)
placement_records: final placement outcomes linked to selected applications

Relationships:

- Company (1) -> (M) PlacementDrive
- Student (1) -> (M) Application
- PlacementDrive (1) -> (M) Application
- Application (1) -> (0/1) PlacementRecord
- Student (1) -> (M) PlacementRecord
- Company (1) -> (M) PlacementRecord

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6. Features

Authentication and RBAC:
- Admin login (predefined; no registration)
- Student and Company registration + login
- Role-based route protection using custom decorators

Admin Module:
- Dashboard with totals (students, companies, drives, applications, placements)
- Approve/reject companies
- Approve/reject drives
- Search/manage students and companies
- Activate/deactivate and blacklist operations
- View applications and placement records

Company Module:
- Profile management
- Create/edit/close/delete drives (with approval workflow)
- View applicant list per drive with student details and resume
- Update status: Shortlisted / Accepted(Selected) / Rejected

Student Module:
- Register/login and update profile
- Mandatory resume upload during registration
- CGPA validation range: 0 to 10
- View/search approved drives
- Apply from drives page and dashboard
- Track application status and placement history

Validation and UX:
- HTML5 form validation (required, type, min/max, pattern)
- Backend validation in Flask routes
- Flash message feedback with auto-dismiss behavior
- Bootstrap responsive layout for desktop/tablet/mobile

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7. Video Presentation

Drive Link:
https://drive.google.com/file/d/1YaboIxoef3WXMmpkxFSbFKDLeb-r1xC5/view?usp=sharing
(Accessible to all with "View" permission.)